

HARIS WAHEED BHATTI

Nationality: Pakistani | Date of birth: 25/12/2001

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Work Authorisation: Pakistani national residing in Slovenia on a temporary student residence permit with full-time work rights; eligible for a post-study job-search residence permit from September 2026.

PROFILE

BIM and digital construction specialist completing a European Master's in Building Information Modelling at the University of Ljubljana, building on a Bachelor's in Civil Engineering from NUST, Pakistan. Experienced across IFC-to-graph-database pipelines and blockchain-based construction document management. Seeking roles such as **Information Manager**, **BIM Specialist**, **AI in BIM**, or **IFC Expert** in Slovenia.

EDUCATION

European Master in Building Information Modelling (BIM A+) <i>University of Ljubljana, UL FGG</i> - Field of study: Information Management in Construction Thesis: <i>Graph-Mediated LLM Exploration for IFC Question Answering: A Benchmark Comparison Against Direct-IFC Approaches.</i> Supervisor: Prof. Žiga Turk	10/2024 – 07/2026 Slovenia
BE Civil Engineering <i>National University of Science and Technology</i> - Final grade: 3.25/4.00 Thesis: <i>Enhancing Construction Transparency and Dispute Resolution Through Blockchain-Based BIM Models</i> – developed a Solidity smart contract to digitally sign and manage construction documents, with off-chain storage linked to the blockchain for immutability and transparency - Funded by a PKR 100,000 grant from the Pakistan Engineering Council; designed to streamline dispute resolution for construction stakeholders - Link: bit.ly/49dUTmK Contract: sepolia.arbiscan.io/address/0x8535...c867	09/2020 – 06/2024 Pakistan

PROJECTS

BIMGraphExplorer <i>Neo4j, Cypher, Python, IfcOpenShell, MCP</i> - Published <code>ifctoneo4j</code>, a Python package converting IFC (IFC2X3, IFC4, IFC4X3) into Neo4j property graphs under the Linked Building Data ontologies with no RDF intermediate - Built an adaptive querying system exposing four read-only tools over the Model Context Protocol, in which the model profiles the graph schema and issues iterative Cypher queries conditioned on prior results - Ran a controlled benchmark against a published direct-IFC baseline at matched model, matched validated judge, and matched scope of 23 IFC datasets (367 questions, 15 projects): 72.0 % versus 82.3 % accuracy, transparency at parity (99.7 % versus 100 %), and outright wins on 5 of 23 datasets - Raised accuracy by 33 percentage points on a fixed model and fixed dataset through prompt design alone, lifting source-transparency from 12.5 % to 99.7 %	2026
Pakistani-Themed Expo BIM Management <i>IFC, Autodesk Construction Cloud</i> - Spearheaded a BIM management project for a Pakistani-themed expo in Guimarães, Portugal, leveraging IFC and Autodesk Construction Cloud to develop a comprehensive technical design - Facilitated collaboration between Architecture and MEP disciplines, creating a federated model that integrated diverse design inputs - Conducted clash detection and compliance checks to ensure the model met all information requirements and design standards	28/10/2024 – 19/11/2024

EXPERIENCE

Contract Management Intern <i>Engineering and Management Services</i> Reviewed construction contract reports and arbitration case documentation under FIDIC conditions, and presented findings on FIDIC and arbitration procedures to the internal legal team	08/2023 – 09/2023 Pakistan
Research Intern (Part-time, Remote – Concurrent with Above) <i>Southern Illinois University Edwardsville</i> - Extracted and synthesized findings from academic articles and abstracts to support a faculty-mentored research proposal, applying feedback to refine presentation and analysis technique - Supervisor: linkedin.com/in/saadullah1	07/2023 – 08/2023 USA
Civil Engineering Intern <i>Fatima Fertilizer Company</i> Modeled a 42m x 162m (6,804 m²) rock phosphate storage shed independently in 3D from 2D drawings, coordinating with stakeholders to refine the design	07/2023 – 08/2023 Pakistan

SKILLS

BIM & AEC: IFC, IfcOpenShell, Revit, Dynamo, Blender, Autodesk Construction Cloud, Three.js, WebIFC
Programming & Data: Python, TypeScript, Solidity, Neo4j, Cypher, knowledge graphs, Git, LaTeX
AI / LLM: agentic systems, Claude Agent SDK, MCP, LLM evaluation and benchmarking
Languages: Urdu (native), English (fluent)

LEADERSHIP & ACTIVITIES

Leadership & Extracurricular Activities
National University of Science and Technology

2020 – 2024
Pakistan

- **Director, School Orientation Committee:** Coordinated logistics and onboarding programming for incoming student cohorts during NUST Orientation
- **President, NUST 3v3 Madness Basketball Tournament:** Organized team registration and event logistics for a university-wide competition

CERTIFICATIONS

Coursera Certificates

Supervised Machine Learning: Regression and Classification , DeepLearning.AI Verify	19/02/2023
Management of Urban Infrastructures – Part 1 , EPFL Verify	15/02/2024
Innovative Governance of Large Urban Systems , EPFL Verify	21/02/2024